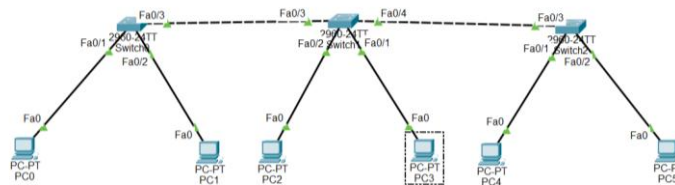
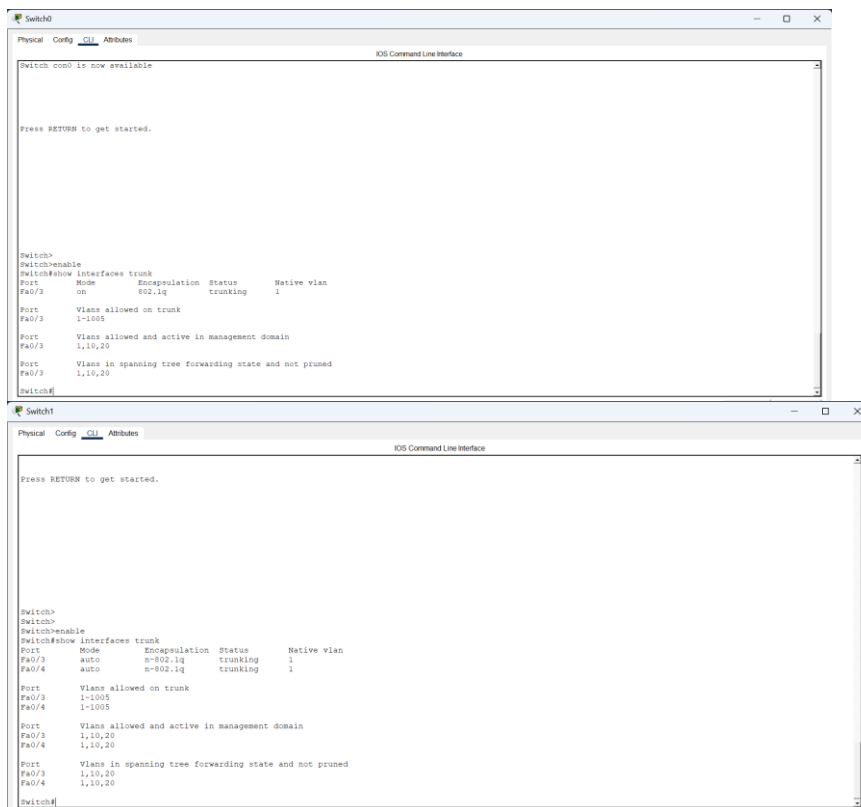


Tasks'

1. On Cisco Packet Tracer, create a network with three switches and assign at least two different VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support'). Connect two PCs to each switch and assign each PC to the correct VLAN based on department.



2. Configure trunk ports between the switches to allow VLAN traffic to pass through. Verify trunking by using the show interfaces trunk command and submit a screenshot of the output.



```
Switch2>enable
Switch2#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/3     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/3     1-1005

Port      Vlans allowed and active in management domain
Fa0/3     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/3     1,10,20

Switch2#
```

3. Set up a Layer 3 device (Router-on-a-Stick using a router or a multilayer switch) to enable inter-VLAN routing between VLAN 10 and VLAN 20. Assign appropriate sub-interfaces and IP addresses for each VLAN.



```
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#
Router(config)#
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
Router(config)#
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#end
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned     YES unset  up          up
GigabitEthernet0/0.10  192.168.10.1   YES manual up          up
GigabitEthernet0/0.20  192.168.20.1   YES manual up          up
GigabitEthernet0/1    unassigned     YES unset  administratively down down
Vlan1           unassigned     YES unset  administratively down down
Router#
```

4. Test connectivity by pinging between a PC in VLAN 10 and a PC in VLAN 20. If pings fail, troubleshoot and fix the configuration so that inter-VLAN communication works. Hint: Check sub-interface IPs, trunk configuration, and default gateways on PCs.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=16ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 4ms
C:\>
```